UCC 103 PRINCIPLES OF COMPUTING

CHAPTER EIGHT: INTRODUCTION TO WINDOWS

At the end of the chapter the learner shall be able to;

- Explain the process of booting a computer
- Start a computer system and log on to windows operating system
- Start a program using the all programs menu#
- Turn off and restart a computer
- Retrieve files, create a folder and a shortcut to a program

8.1 Windows operating system Boot up Process

Booting Up

The boot-up process is a list of detailed procedures that the system undergoes to perform all system checks and load all necessary files to bring the computer to an operable state.

It is useful to understand what happens behind the scenes when you switch on your computer from an idle machine to an operable and functional system. There are essentially two forms of booting - the **soft boot** and the **hard boot**. The **warm boot or hard boot** involves powering the computer up from an initial zero power supply. A **cold boot** on the other hand takes place when a software application or operating system triggers the computer to perform a reboot.

A successful boot is dependent on 3 conditions - the hardware, BIOS and operating system files to function without errors. When an error occurs, you will be notified by error messages, beeping sounds or in the worst scenario, a blank screen.

Boot-up Process

The boot-up process is a list of detailed procedures that the system undergoes to perform all system checks and load all necessary files to bring the computer to an operable state.

The boot up sequence of events

The boot-up process is a list of detailed procedures that the system undergoes to perform all system checks and load all necessary files to bring the computer to an operable state. When the computer is switched on, it needs instructions to start. BIOS contain the instructions for the starting up of the computer. The BIOS runs when the computer is switched on. It performs a **Power On Self Test (POST**) that checks that the hardware is functioning properly and the hardware devices are present. It checks whether the operating system is present on the hard drive. BIOS invokes the bootstrap loader to load the operating system into memory. BIOS can be configured using an interface named BIOS setup, which can be accessed when the computer is booting up (by pressing the DEL key).

BIOS setup program, which is a built-in utility in BIOS, lets the user set the many functions that control how the computer works. BIOS displays the system settings and finds the bootable devices. It loads the interrupt handlers and device drivers. It also initializes the registers.

Bootstrap Loader is a program whose purpose is to start the computer software for operation when the power is turned on. It loads the operating system into RAM and launches it. It generally seeks the operating system on the hard disk. The bootstrap loader resides in the ROM. The BIOS initiates the bootstrap sequence.

Chip BIOS ROMs are accompanied by a smaller CMOS (CMOS is a type of memory technology) memory chip. CMOS chip saves some system information, such as time, system date and essential system settings. CMOS is kept powered by a button battery located on the motherboard. The CMOS chip is working even when the computer power is switched off. Information of the hardware installed in the computer (such as the number of tracks or sectors on each hard drive) is stored in the CMOS chip.

The Windows boot-up process comprises of the following procedures:
1) The Power-On Self Test Phase
2) BIOS ROM Phase
3) Boot Loader Phase
4) Operating System Configuration Phase
5) Security & Logon Phase

8.2 Logging On to Windows

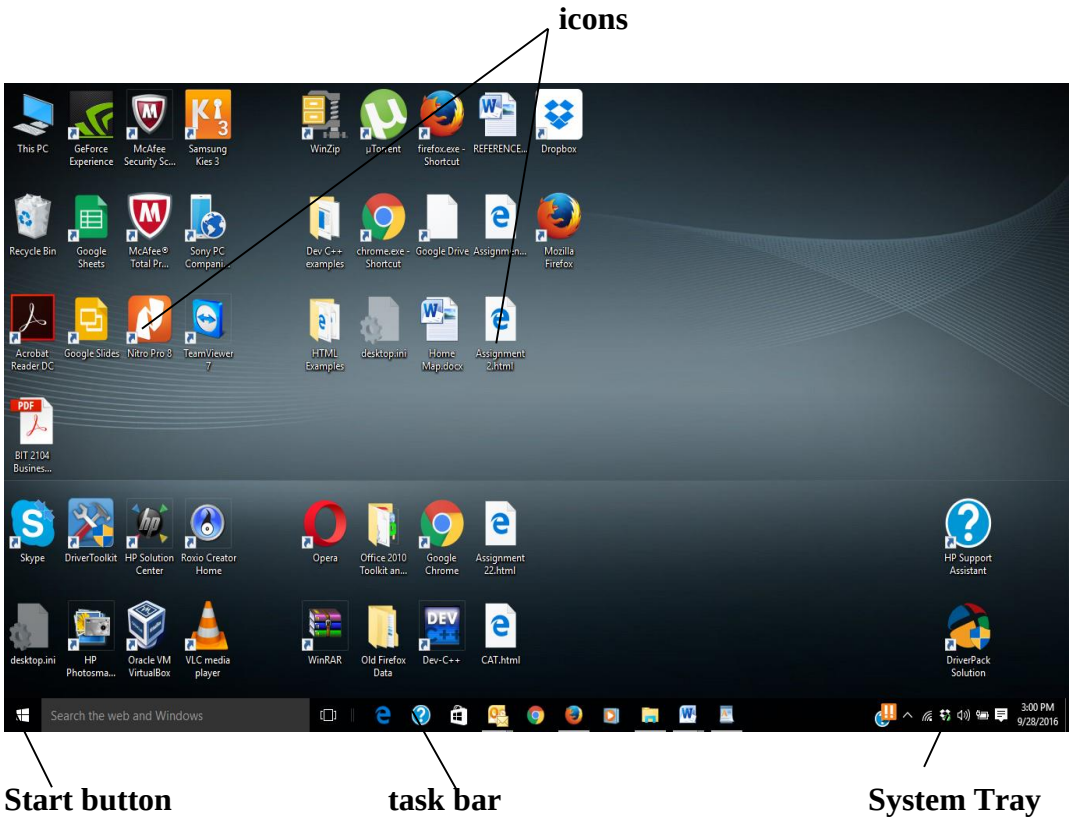
You must log on to the PC before you can use it. Each user is given a user name and a password which must be entered in the log-on screen before Windows will start up. When you log on you will have access to your own personal space.
The default Login screen for Windows computers is appears when you switch on the machine. To login; Enter your **user** and **password** in the respective boxes.

When you enter your password, Windows will display a series of ●●●●●●●●●●●s to protect your password from wandering eyes.
Press ***ENTER*** on the keyboard or click ***OK***

8.3 Windows Desktop

Desktop is a system of organization of icons on a screen. It is your work area on which program and files icons are located. It is basically a workspace where you can access everything you need to operate your computer, such as system components, applications, the Internet, etc.

The desktop contains:

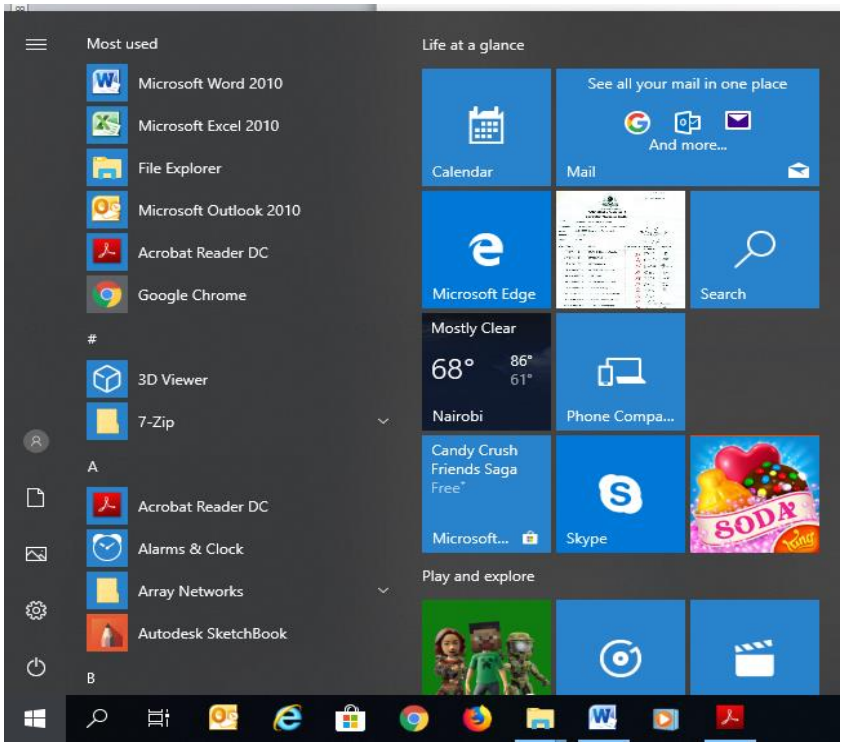


The desktop contains:

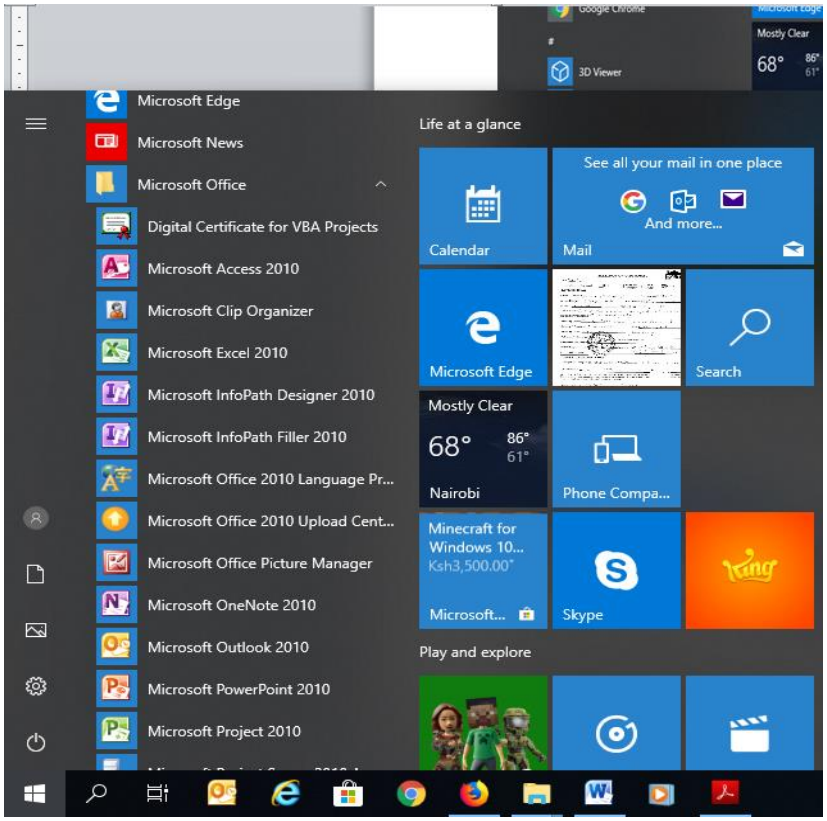
1) **Start button**: one of the most important tools you will use while working with Windows. The Start button allows you to open menus and start applications.
2) **Taskbar**: primarily used to switch between open windows and applications
3) **Icons (or graphical pictures)**: represent applications, files, folders and other parts of the operating system. By default Windows provides you with one desktop icon, the Recycle Bin. Learn more about the Recycle Bin in a later module.
4) **My Computer.** The My Computer icon provides access to the resources on your computer. You can access your drives and other peripherals by clicking on the My Computer icon. You can also access the Control Panel through My Computer.
5) **Internet Explorer.** The Internet Explorer icon launches the Internet Explorer browser.
6) **My network places.** If you are working on a network, this displays all of the computers on the network
7) **System Tray**. The part of the taskbar that holds the clock, volume control, and icons for other utilities that runs in the background of your system.
8) **Recycle Bin**. This is where you dump any files you want to delete.

8.4 Starting a Program

1. Click the **Start button**, programs menu will popup.



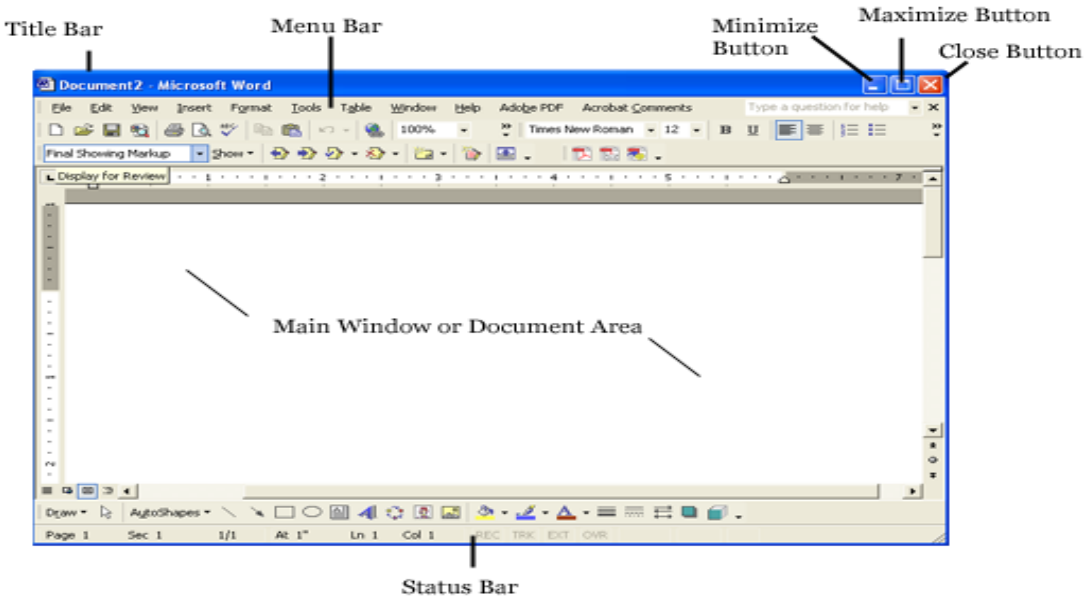
3. **Point the cursor** to the required Program Name and click.
A sub-menu window will popup is the program has associated sub-programs as shown below.



4. Click on the required program name for the program to open

8.5. Parts of a Window

Windows contain buttons and menus to control the program and window. Windows are used in most programs, but the good news is once you learn the window of one program, you will be familiar with the windows for most programs since the window, menu and button layout appears in just about every window program.



1) **Close Button** Closes the window or program, removing it from the screen and the computer's memory.
2) **Minimize Button** Minimizes a program from view. The program is still ready for use and can be found in the task bar.
3) **Maximize Button** Enlarges the window so that it fills the entire screen, allowing you to see your entire workspace. Notice, when a window is maximized, this button is not shown.
4) **Restore Button** When a window is maximized, this button is shown. Clicking it will make the window smaller.
5) **Menu Bar** Controls what a program does. The menu functions listed will change from program to program, but the bar is always located at the top.
6) **Status Bar** Displays information about the program, such as instructions or special information.
7) **Title Bar** Displays the name of the program and name of the file in use.
8) **Main Window** .This is where you work within a program. If it is Microsoft Word, this is where you would type, if it is a web browser this is where the web page would be displayed.
9) **Toolbar.** Have shortcuts to the menu items

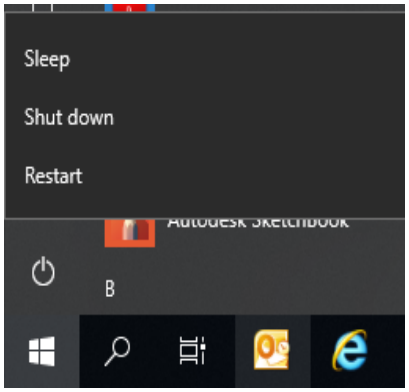
8.6. System Management: Log off, Switch Users, Restart, Lock System

More than one person may use your computer. For example several coworkers may be able to access your computer on a computer network. Windows allows everyone who uses your computer

to have separate computer accounts. A computer accounts tracks each person's unique settings, documents, and email accounts.
Windows even enables you to log off the computer so someone else can log on without having to restart the computer.

8.6.1. Login off

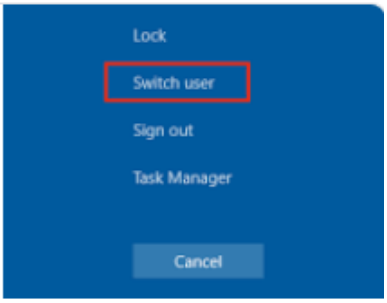
1) Click on start button
2) Click on power button



3. Click the appropriate menu option
   a) **Sleep:-** System is put to sleep where it slows down its activities, power usage etc. sometimes a user password may be required to "wake" it up
   b) **Shut down**: All applications started will be closed and the system will be shut down/powered off
   c) **Restart**: the system shuts down and restarts again (re-boots)

8.6.2. Change User, Lock and Access Task Manager

1) Click Ctrl +Alt + Delete button and the same time



2) Select and Click User to select another user
3) Select and Click Lock to lock the system
4) Click and Select Sign-out to Log-off
5) Click and select Task Managers to open the Task Managers window

8.7 Working with Files and Folders

Computer folders are named memory spaces that can be used to store files.

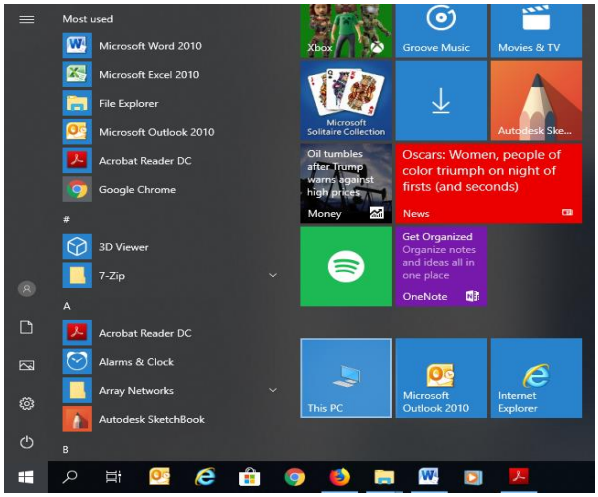
8.7.1. Creating a Folder

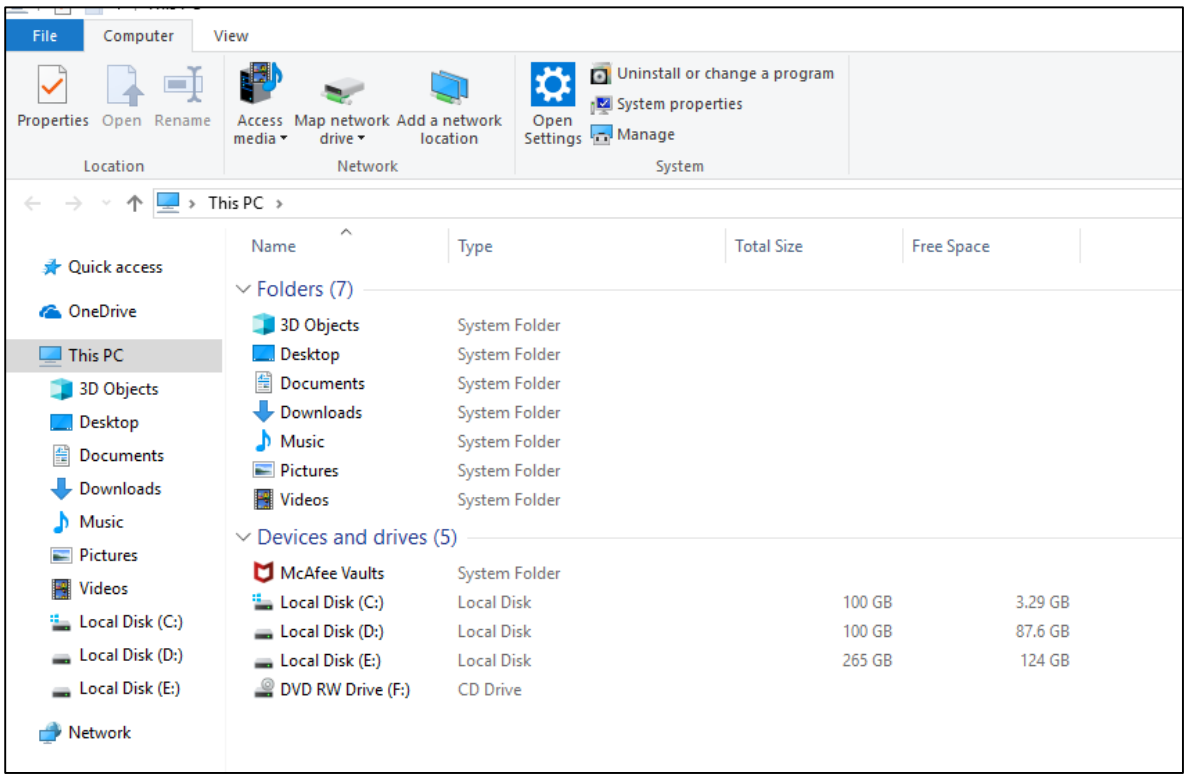
Folders can be create in hierarchical tree format where folders can hold sub-folders etc.

Folder A
└— Folder B (Sub-folder of A)
└— Folder C (Sub-folder of B)

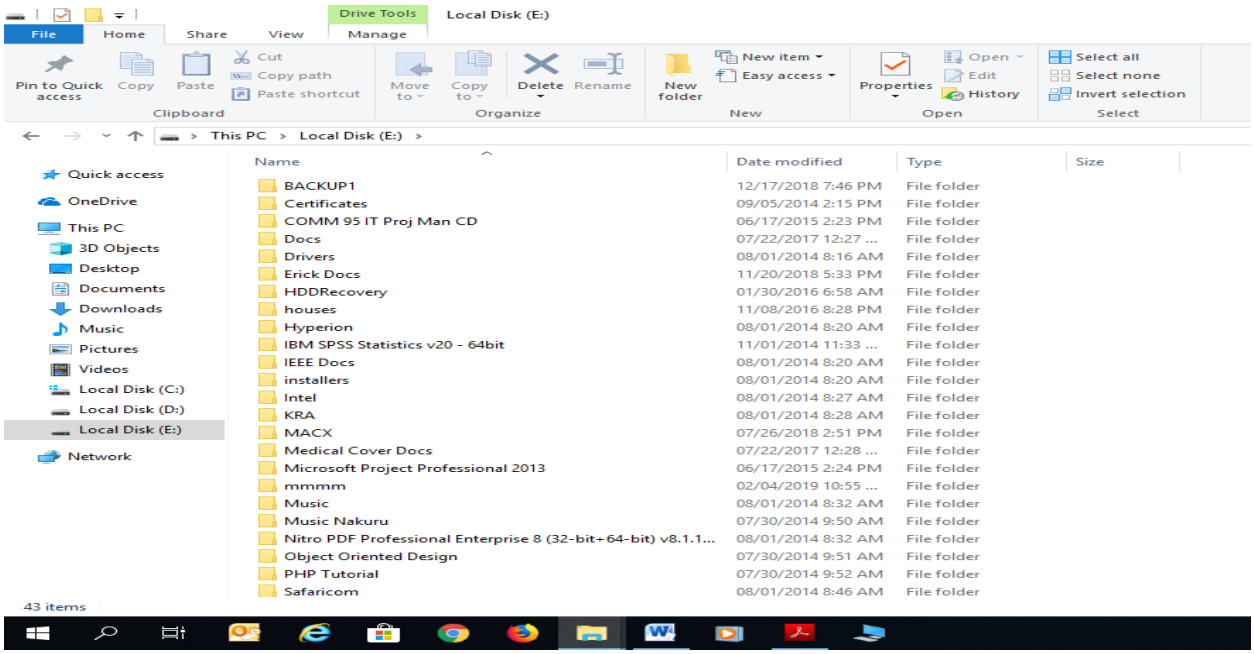
Create a folder on the C-Drive

1) Click Start button
2) Select and Click This PC (My Computer – in older versions of Windows)





3) Double Click the Appropriate Disk to open it



4) Select and click New folder option on the Menu Bar. The folder will appear at the bottom of your list. When it displays as a black rectangle with the words **New Folder** highlighted in blue, it is prompting you to give it a name.
5) Type in the name of the folder created and press Enter Key

8.7.2. To Create a folder in side another folder – Sub-folders.

1) Double Click the proposed parent folder and open it
2) Select and click New folder option on the Menu Bar. The folder will appear at the bottom of your list. When it displays as a black rectangle with the words **New Folder** highlighted in blue, it is prompting you to give it a name.
3) Type in the name of the folder created and press Enter Key

8.7.3. Open your Folders

1) Double click to open the Local Disk where the folder is located
2) Select and Double click on the folder.
3) The contents of the folder will be displayed in the open window. To get back to the previous folder, click the Back Button.

8.7.4. Renaming a Folder

1) Select the folder
2) Right Click. A menu will appear. Close to the bottom you will see the word **Rename**.
3) Click Rename me menu and a black rectangle with the word highlighted in blue will appear. Rename your folder and hit **Enter**.

OR

1) Select the folder

2) Select and Click Rename button in the Organise menu

8.7.5. Deleting a folder

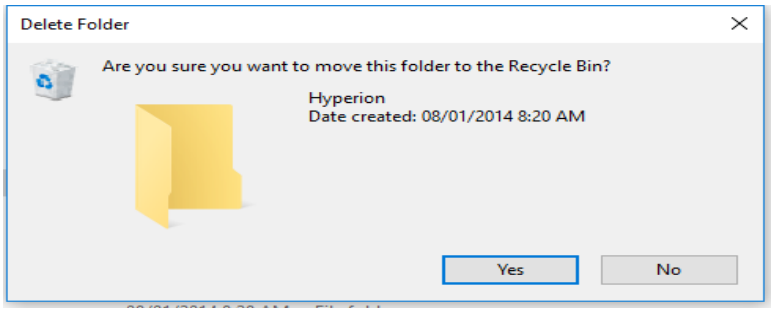
1) Select the folder and Right click.
2) Select Delete button in the subsequent menu

OR

Select the folder

1) Click the Delete button in the Organize menu
2) Select the file, go to file menu and select delete

A dialog box appears asking if you want to delete the folder, click yes



Once folders or files are deleted they go to the recycle bin which is a holding or storage location for files not required. The still remain in the computer hard disk. These files can be retrieved from recycle bin as long as recycle bin have not been emptied.

8.8. Copying files in folders

Copying file between folders allows you to have two duplicate copies of the file in different folders. A copy of the file is left in the Source folder while a duplicate copy is maintained in any subsequent folder it is copied to.

8.8.1. Using Menu option

Source Folder

1) Select the file from the Source folder
2) Select and Click Clipboard menu
3) Select and click copy

OR

1) Select and Click Organize menu
2) Select and Click Copy To

Destination folder

1) Select and double click the Destination folder
2) Select and Click Clipboard menu
3) Select and click Paste

OR

1) Select and click Organize menu

2) Select and Click Paste

8.8.2. By dragging

Open the source and destination folders and place them next to each other

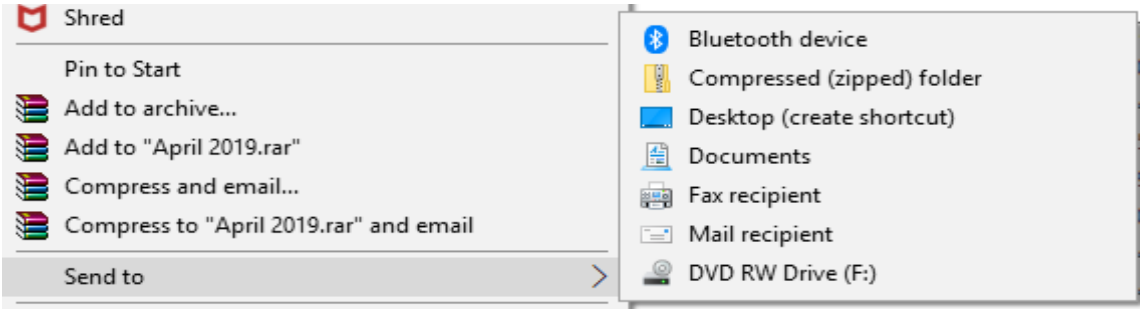
1) Click on the file on source folder and hold the mouse left button down
2) Drag it to the destination folder and release the button

8.9. Short cuts

Short cuts are icons placed on the Desktop that allows you to quickly open a program or file

Procedure of creating shortcuts

1) Locate the item and right click
2) Select Send to menu
3) Select Desktop (create shortcut)



8.10. Recycle Bin

Windows provides us with a folder called recycle bin where all deleted files are stored. If you accidently delete a file you can be able to retrieve it from the recycle bin and restore it to the folder it originally was stored in. The Recycle bin icon is always place on the desktop.

The folder is the equivalent of dust bin in an office.

Retrieving files

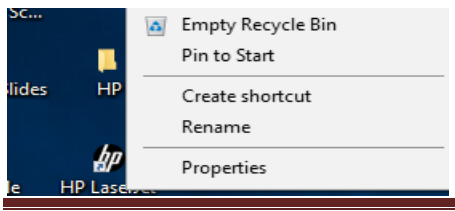
1) Select and double click the recycle bin Icon on the Desktop to open it **OR** Select, Right click and select Open
2) Locate the file and right click on it and select restore.

Emptying Recycle Bin

3) Select and double click the recycle bin Icon on the Desktop to open
4) Select and click Empty Recycle Bin from the Manage Menu

OR

1) Right Click the Recycle Bin Icon
2) Select Empty Recycle Bin



8.11. Memory Formatting

A format is a pre-established layout for data. A computer program accepts data as input in a certain format, processes it, and provides it as output in the same or another format. All data is stored in some format with the expectation that it will be processed by a program that knows how to handle that format. Generically data formats tend to fall into bitmaps (strings of 0s and 1s) that describe images or sound patterns (or both), text formats (in which usually each byte value is mapped to character) and numeric data formats (used by spreadsheet and other database programs)

Disk formatting is the configuring process of a data storage media such as hard disk drive, floppy disk or flash drive for initial usage. Any existing files on the drive would be erased with disk formatting. Disk formatting is usually done before initial installation or before installation of a new operating system. It is also done if there is a requirement for additional storage in the computer. Disk formatting can be performed on both magnetic platter hard drives and solid-state drives. The formatting comprises low-level formatting, partitioning and high level formatting. Low- level formatting aids in preparing the physical structure on the storage media. The partitioning process involves the division of the hard drive into logical volumes for data storage. High- level formatting helps in creating the file system format within the logical volume or within the disk partition. Disk formatting is usually done with the help of a disk formatting utility.

While preparing the hard drive for initial use, disk formatting checks for errors in the drive. It can scan and repair bad sectors. Another benefit associated with disk formatting is its capability to erase bad applications and remove sophisticated viruses. Disk formatting is an action which must be done with caution. As it deletes data and removes programs installed, backup of the necessary data or applications are required. Disk formatting takes time. Frequent disk formatting can gradually decrease the life of a hard drive.

8.12. **Chapter Review Questions**

1. Which of the following is not contained in the Windows desktop
(a) Recycle Bin (b) Task Bar (c) Start Button (d) File menu

2. Which of the following has the command for turning off the computers system
(a) Recycle Bin (b) Task Bar (c) Start Button (d) File menu

3. A user can create a folder on the desktop. True or false

4. Which of the following is not part of a window?
(a) Task Bar (b) Menu Bar (c) Title Bar (d) Close Button

5. Explain the steps you would follow to create a folder in my documents location.